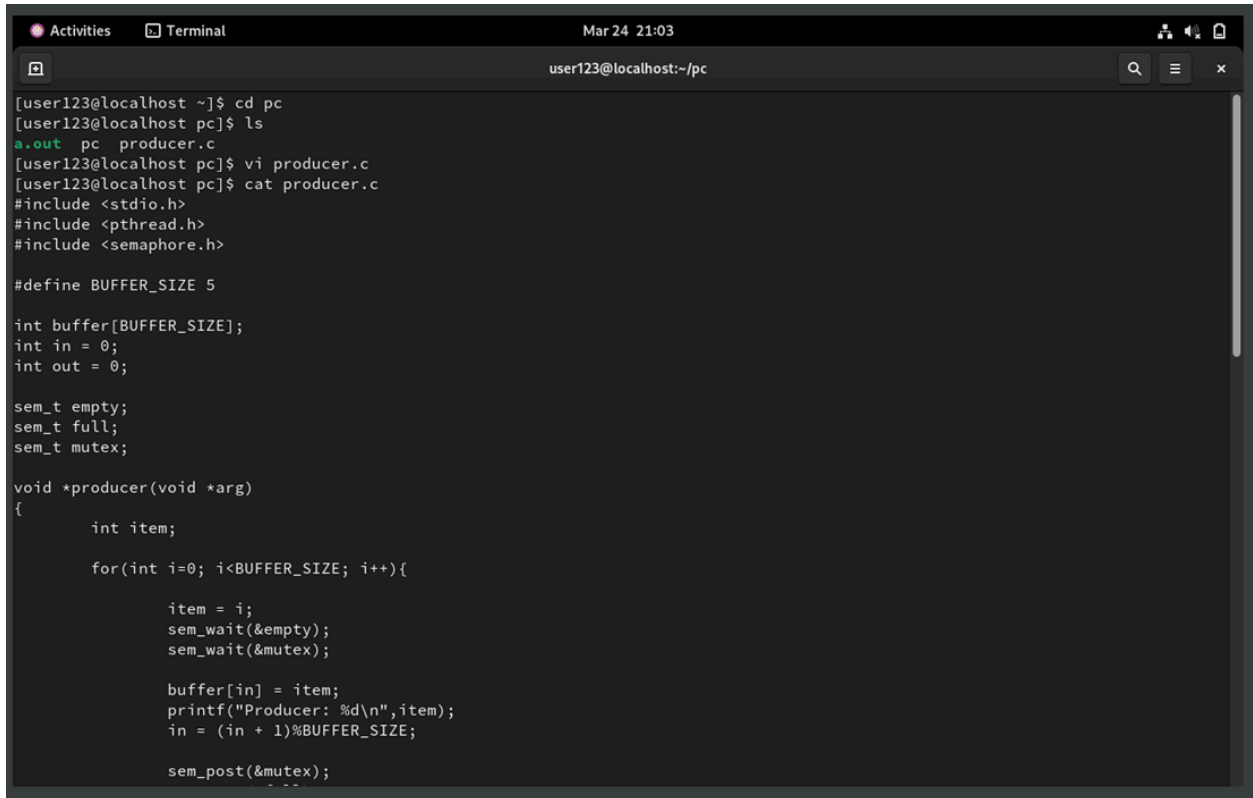


241401084

PRODUCER CONSUMER PROBLEM SOLUTION USING SEMAPHORE



```
Activities Terminal Mar 24 21:03 user123@localhost:~/pc
[user123@localhost ~]$ cd pc
[user123@localhost pc]$ ls
a.out pc producer.c
[user123@localhost pc]$ vi producer.c
[user123@localhost pc]$ cat producer.c
#include <stdio.h>
#include <pthread.h>
#include <semaphore.h>

#define BUFFER_SIZE 5

int buffer[BUFFER_SIZE];
int in = 0;
int out = 0;

sem_t empty;
sem_t full;
sem_t mutex;

void *producer(void *arg)
{
    int item;

    for(int i=0; i<BUFFER_SIZE; i++){

        item = i;
        sem_wait(&empty);
        sem_wait(&mutex);

        buffer[in] = item;
        printf("Producer: %d\n", item);
        in = (in + 1)%BUFFER_SIZE;

        sem_post(&mutex);
    }
}
```

```
Activities Terminal Mar 24 21:04 user123@localhost:~/pc
sem_post(&mutex);
sem_post(&full);
}
}

void *consumer(void *arg)
{
    int item;

    for(int i=0; i<BUFFER_SIZE; i++) {
        sem_wait(&full);
        sem_wait(&mutex);

        item = buffer[out];
        printf("Consumed: %d\n", item);
        out = (out + 1) % BUFFER_SIZE;

        sem_post(&mutex);
        sem_post(&empty);
    }
}

int main()
{
    pthread_t p, c;

    sem_init(&empty, 0, BUFFER_SIZE);
    sem_init(&full, 0, 0);
    sem_init(&mutex, 0, 1);

    pthread_create(&p, NULL, producer, NULL);
    pthread_create(&c, NULL, consumer, NULL);
}
```

```
Activities Terminal Mar 24 21:06 user123@localhost:~/pc
int main()
{
    pthread_t p, c;

    sem_init(&empty, 0, BUFFER_SIZE);
    sem_init(&full, 0, 0);
    sem_init(&mutex, 0, 1);

    pthread_create(&p, NULL, producer, NULL);
    pthread_create(&c, NULL, consumer, NULL);

    pthread_join(p, NULL);
    pthread_join(c, NULL);

    sem_destroy(&empty);
    sem_destroy(&full);
    sem_destroy(&mutex);

    return 0;
}
[user123@localhost pc]$ gcc producer.c
[user123@localhost pc]$ ./a.out
Producer: 0
Producer: 1
Producer: 2
Producer: 3
Producer: 4
Consumed: 0
Consumed: 1
Consumed: 2
Consumed: 3
Consumed: 4
[user123@localhost pc]$
```

